

QuizMaster Application

Business Requirement Document (BRD)

This epic focuses on developing an interactive, console-based Java Quiz System that enables students to register, log in, and attempt Java-related quizzes through a terminal interface. The system is designed using Core Java, JDBC, and MySQL, and emphasizes a seamless user experience entirely via the console.

The application includes robust user management, allowing students to securely register and log in using validated credentials. Upon successful authentication, users can participate in a multiple-choice quiz, after which their score is automatically calculated and stored in the database. Each student receives a score and performance grade based on predefined criteria.

In parallel, the admin module provides essential features to manage quiz content and monitor student performance. Admins can add, update, and delete quiz questions, as well as view all student scores, retrieve scores by student ID, and identify top performers.

The system ensures input validation, exception handling, and adherence to Java best practices (OOP design, modular methods, and code readability). It also maintains data integrity using relational constraints and secure data access via JDBC

Technical Requirements

- Language: Core Java
- Database: MySQL
- Access: JDBC
- UI: Console-based
- Validation: Email, Mobile number, unique usernames
- Design:
 - OOP concepts used (Class for Student, Admin, Quiz)
 - Methods separated for registration, login, question handling, score calculation
 - Exception handling for DB connectivity and input mismatches
 - Collections used where needed (e.g., storing temporary quiz answers)

Epic 1: User Management

User Story 1.1: Student Registration

As a student I want to register by providing personal details
So that I can log in and access the quiz

Acceptance Criteria:

Input fields: First Name, Last Name, Username, Password, City, Email, Mobile Number

All fields are validated (e.g., email format, password length, no duplicates)

Data stored in MySQL DB

User Story 1.2: Student Login

As a student I want to log in with my username and password So that I can access the quiz

Acceptance Criteria:

Validates username and password from DB

Displays error for invalid credentials

Epic 2: Quiz Management

User Story 2.1: Display Quiz Questions

As a student I want to view multiple-choice questions So that I can attempt the quiz

Acceptance Criteria:

Display 10 pre-stored Java questions

Each question has 4 options

Answers taken via console input

Invalid option (not 1-4) shows error

User Story 2.2: Score Calculation and Grade

As a student I want to receive a score and grade after the quiz

So that I know how I performed

Acceptance Criteria:

Each correct answer gives 1 point

Final score and grade (A/B/C) shown after quiz

Score stored in DB with student ID

User Story 2.3: View My Score

As a student I want to view my quiz score

So that I can know my performance anytime

Acceptance Criteria:

Student provides username/password

Score is fetched from DB and displayed

Epic 3: Admin Management

User Story 3.1: Add New Questions

As an admin I want to add new questions and answers to the database So that I can keep the quiz up to date

Acceptance Criteria:

Input: Question Text, 4 options, correct answer

Stored in questions table in DB

User Story 3.2: View All Student Scores

As an admin I want to view all student scores So that I can analyze quiz results.

Acceptance Criteria:

Fetch and display all student scores sorted in ascending order.

User Story 3.3: Search Student Score by ID

As an admin I want to search for a student's score by ID So that I can track individual performance

Acceptance Criteria:

Student ID is input

Corresponding score is displayed.

User Story 4.1: Validate User Inputs

As a system I want to validate all user inputs

So that I can avoid invalid or harmful data

Acceptance Criteria:

Email format must be valid (abc@xyz.com)

Password must have a minimum of 6 characters

Mobile must be exactly 10 digits

No field should be empty during registration or login

User Story 4.2: Prevent Duplicate Usernames

As a system I want to ensure no two users can have the same username So that I maintain data uniqueness and login integrity

Acceptance Criteria:

During registration, check DB for existing username

Show appropriate message on conflict.

Epic 5: Result Analysis & Reporting

User Story 5.1: Display Grade with Message

As a student I want to see a message based on my score So that I can get feedback on my performance

Acceptance Criteria:

If score $\geq$ 8: Grade A – "Excellent"

If score $\geq$ 5: Grade B – "Good"

If score $<$ 5: Grade C – "Needs Improvement"

User Story 5.2: View Top Scorer

As a student/admin I want to view the highest scoring student So that I can identify top performance

Acceptance Criteria:

Query to fetch max score

Display student name(s) with top score

Epic 6: Question Handling Improvements

User Story 6.1: Prevent Duplicate Questions

As an admin I want to prevent adding the same question again So that I avoid redundancy

Acceptance Criteria:

Question text is compared with existing entries

If match found, show message and skip insertion

User Story 6.2: Edit Existing Question

As an admin I want to edit any existing question and its options So that I can fix mistakes

Acceptance Criteria:

Input: Question ID

Admin can update text and/or options

User Story 6.3: Delete a Question

As an admin I want to delete any existing question So that I can manage irrelevant content

Acceptance Criteria:

Input: Question ID

Question and options are removed from DB

Epic 7: Quiz Functionality Enhancements

User Story 7.1: Shuffle Questions

As a student I want to get questions in random order So that every quiz session is unique

Acceptance Criteria:

Retrieve and display questions randomly using SQL or Java collections

User Story 7.2: Display Quiz Summary

As a student I want to see a summary after completing the quiz So that I can review my attempt

Acceptance Criteria:

Summary includes: Total Questions, Correct Answers, Wrong Answers, Score,

Grade

Epic 8: System & Usability Enhancements

User Story 8.1: Exit and Retry Option

As a user I want to be able to exit or retry any operation So that I can control navigation

Acceptance Criteria:

After each menu or failure, ask user: "Do you want to continue? (Y/N)"

Database Schema Highlights

Tables

student(id, first_name, last_name, username, password, city, email, mobile)

question(id, question_text, option1, option2, option3, option4, correct_option)

score(student_id, total_score, grade)

Foreign key: score.student_id → student.id

User Story Details

Welcome to Quiz based application

User Operations:

1. Register with personal details (name, username, password, email, city, mobile).
2. Log in using a valid username and password.
3. Attempt a Java-related multiple-choice quiz via console.

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4. View quiz result with score and grade after submission.
 5. Retrieve quiz score anytime using login credentials.

Admin Operations:

1. Add new quiz questions with four options and correct answers.
2. Edit or delete existing questions from the quiz database.
3. View all student scores sorted in ascending order.
4. Fetch individual student scores using student ID.
5. Identify and view top-scoring students.

Output Screen

User Operation

1. Student Registration

Enter First Name:

Enter Last Name:

Enter Username:

Enter Password:

Enter City:

Enter Email ID:

Enter Mobile Number:

2. Student Login

Enter Username:

Enter Password:

3. Attempt Quiz

=== Java Quiz ===

You will be asked 10 multiple-choice questions.

Each question has 4 options. Enter your choice (1-4).

Question 1:

<question text>

1. Option A

2. Option B

3. Option C

4. Option D

Enter your answer (1-4):

(Repeat for each question.)

4. View Quiz Result

=== View Quiz Result ===

Enter Username:

Enter Password:

Your Score: X / 10

Your Grade: A / B / C

Feedback: Excellent / Good / Needs Improvement

Admin Operations

1. Add New Question

Enter Question Text:

Enter Option 1:

Enter Option 2:

Enter Option 3:

Enter Option 4:

Enter Correct Option Number (1-4):

2. Edit Existing Question

Enter Question ID to Edit:

Enter New Question Text:

Enter New Option 1:

Enter New Option 2:

Enter New Option 3:

Enter New Option 4:

Enter New Correct Option Number (1-4):

3. Delete Question

Enter Question ID to Delete:

4. View All Student Scores

=== All Student Scores (Sorted in Ascending Order) ===

Student ID | Name | Score | Grade

5. Fetch Score by Student ID

Enter Student ID:

Score: X

Grade: A / B / C

6. View Top Scorer

Student Name: XYZ

Score: 10 / 10

Grade: A